

Chemical Equilibrium

✚ Forward Reaction

✚ Backward Reaction

✚ Irreversible Reaction

✚ Reversible Reaction

✚ Chemical Equilibrium

↳ Homogeneous Equilibrium

↳ Heterogeneous Equilibrium

✚ Characteristics of chemical equilibrium

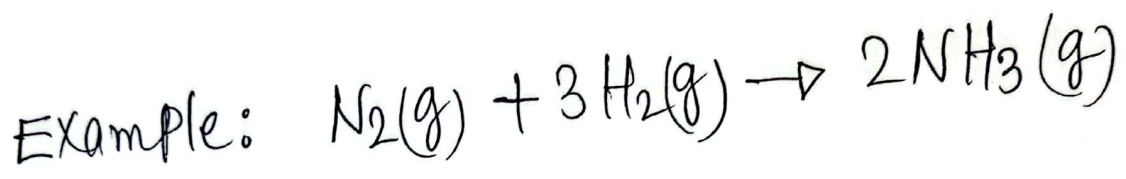
✚ Law of mass action and equilibrium constant.

✚ Relation betⁿ K_p and K_c .

✚ Le-Chatelier Principle.

❖ Forward Reaction: The reaction in which reactants react to form products is called forward reaction.

i.e., $\text{Reactants} \rightarrow \text{Products}$.
 $\rightarrow$ It takes place from left to right.



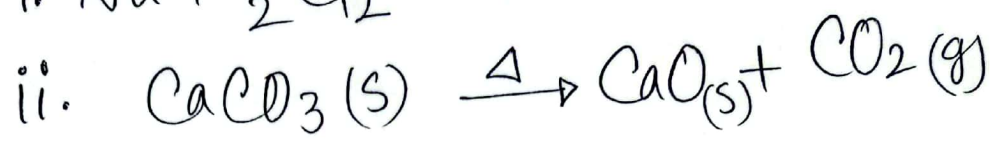
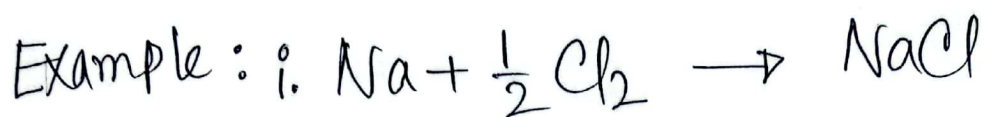
❖ Backward Reaction: The reaction in which products react to produce reactants is called backward reaction.

i.e. $\text{Reactants} \leftarrow \text{Products}$.
 $\rightarrow$ It takes place from right to left.

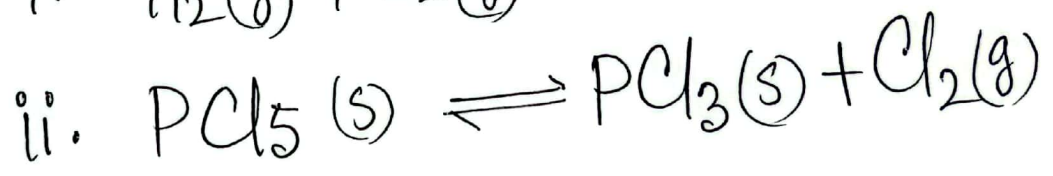
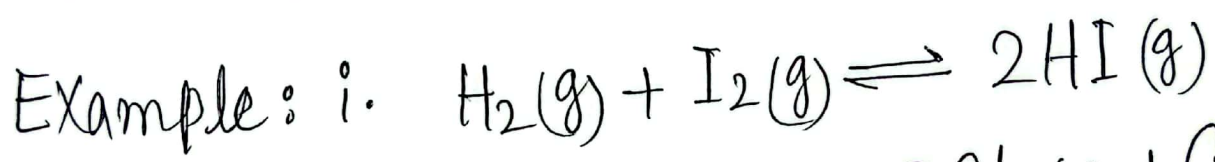


❖ Irreversible Reaction: When in a chemical reaction all the reactants are converted to the products, or in other words the reaction proceeds

in one direction is called irreversible reaction.



Reversible Reaction: A reaction which can go in the forward and backward direction simultaneously is called a reversible reaction.



Chemical Equilibrium:

In case of a reversible reaction the reactants react with one another to produce products. And, on the other hand these products react with each other and form reactants back.

So in these case, when the reaction among the reactants proceeds, after some time the rate of reaction of the reactants to products equals to the rate of reaction of the products to produce the reactants back. The state is called chemical equilibrium.

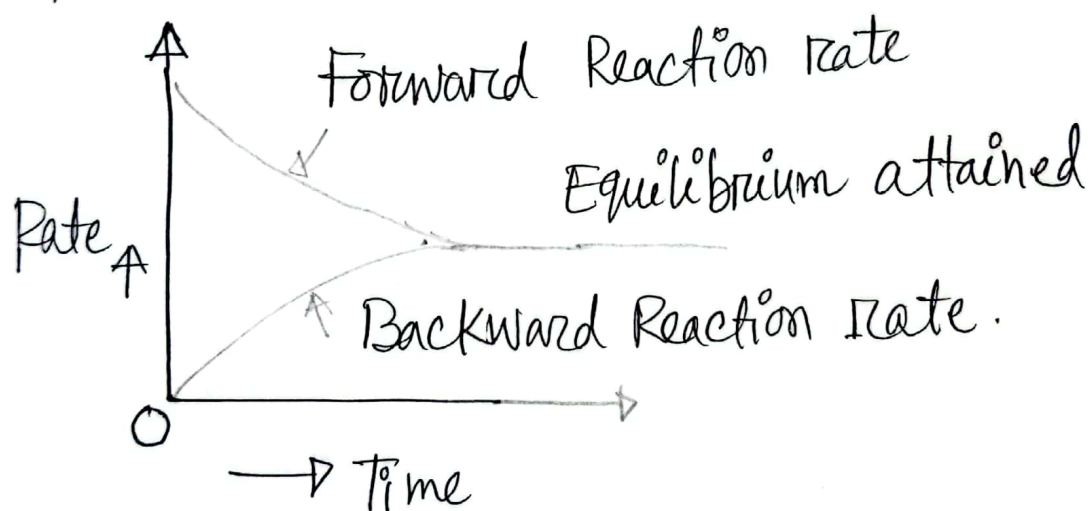
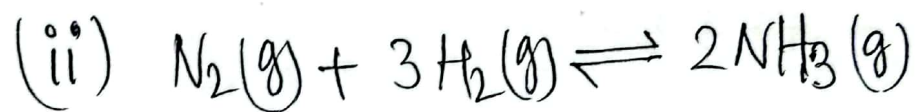
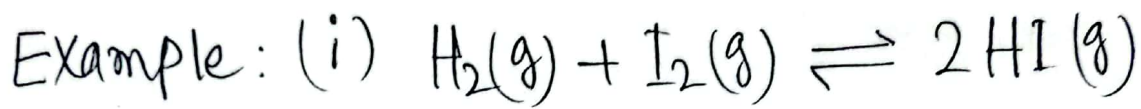


Fig: At equilibrium, the forward & backward reaction rate remains equal.

⇒ There are two types of chemical equilibrium. They are –

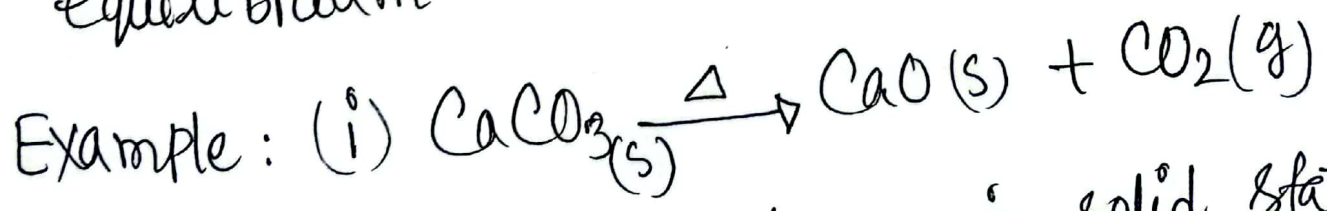
- A. Homogeneous equilibrium (Same state)
- B. Heterogeneous equilibrium (Different state)

A. Homogeneous equilibrium: The chemical equilibrium in which all reactants and products are remain in the same physical state, is called homogeneous equilibrium.



Here, all products and reactants are in gaseous state.

B. Heterogeneous equilibrium: A chemical equilibrium in which at least one product or reactant is in different physical state is called heterogeneous equilibrium.



Here, two components are in solid state and one component is in gaseous state.

⇒ Characteristics of Chemical equilibrium:

There are some characteristics of chemical equilibrium. They are -

1. Reversibility of Reaction: Chemical equilibrium is related only to reversible reactions. It is not applicable to irreversible reaction.
2. Stability of equilibrium: If the external conditions (pressure, temperature and concentration) are not changed, the system will stay for ever.
3. Easy approachability from both sides:
The equilibrium can be attained from both directions.
4. Closed system: It is only attained in closed system.
5. Ineffectiveness of Catalysts: The Catalysts have no effect on chemical equilibrium.

Law of mass action:

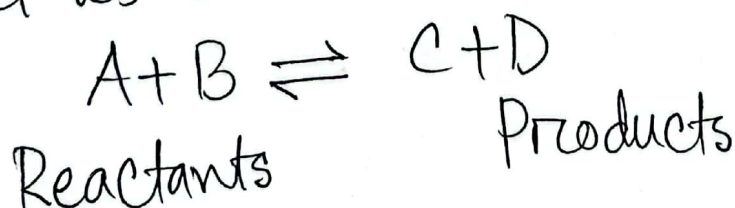
In 1867 Mathematician C.M. Guldberg and Chemist P. Waage of Norway invented a law regarding the rate of chemical reactions, which became known as law of mass action.

The law is as follows:

"The rate of a chemical reaction is proportional to the active masses (i.e. Concentration and Partial Pressure) of the reactants.

⇒ Equilibrium Constant:

Let us consider following general reaction:



According to the law of mass action, the rate of forward reaction,

$$R_f \propto [A][B]$$

$$R_f = k_1 [A][B]$$

Where, $[A]$ & $[B]$ are the molar concentration of the reactants A & B respectively.

K_1 = Rate constant of the forward reaction

Similarly, the rate of backward reaction,

$$R_b \propto [C][D]$$

$$R_b = K_2 [C][D]$$

At the equilibrium, the rate of forward reaction and backward reaction remain same.

$$R_f = R_b$$

$$K_1 [A][B] = K_2 [C][D]$$

$$\Rightarrow \frac{K_1}{K_2} = \frac{[C][D]}{[A][B]}$$

$$\therefore K_c = \frac{[C][D]}{[A][B]}$$

Where,

$[C]$ = Molar concentration of the product C.

$[D]$ = Molar concentration of product D.

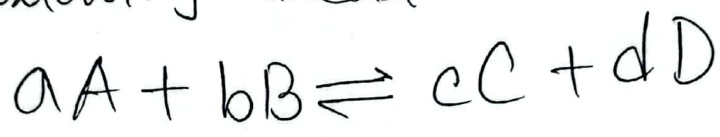
K_2 = Rate constant of the backward reaction.

Here, K_c is called molar equilibrium constant or simply equilibrium constant of the reaction.

$$\therefore K_c = \frac{[C][D]}{[A][B]}$$

This is the mathematical expression for the law of chemical equilibrium.

* Write the K_c & K_p value for the following reaction.



Forward Reaction Rate,

$$R_f \propto [A]^a \times [B]^b$$

$$\therefore R_f = K_1 [A]^a [B]^b$$

Backward Reaction Rate,

$$R_b \propto [C]^c [D]^d =$$

$$R_b = K_2 [C]^c [D]^d$$

At equilibrium,

$$R_f = R_b$$

$$K_1 [A]^a [B]^b = K_2 [C]^c [D]^d$$

$$\text{or, } \frac{K_1}{K_2} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\text{or, } K_c = \frac{C_C^c \times C_D^d}{C_A^a \times C_B^b}$$

$$\text{And, } K_p = \frac{P_C^c \times P_D^d}{P_A^a \times P_B^b}$$

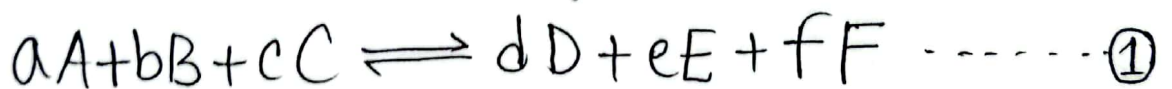
Where, K_c = Equilibrium constant at molar concentration

K_p = Equilibrium constant at Partial Pressure.

✱ Establish the relation between K_p and K_c for the given reaction.



Let us consider the given reversible reaction,



The rate of forward reaction,

$$R_f \propto [A]^a [B]^b [C]^c$$

$$R_f = K_1 [A]^a [B]^b [C]^c \text{ ----- ②}$$

The rate of backward reaction,

$$R_b \propto [D]^d [E]^e [F]^f$$

$$R_b = K_2 [D]^d [E]^e [F]^f \text{ ----- ③}$$

At equilibrium,

$$R_f = R_b$$

$$\text{or, } K_1 [A]^a [B]^b [C]^c = K_2 [D]^d [E]^e [F]^f$$

$$\text{or, } \frac{K_1}{K_2} = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c}$$

$$\text{or, } K_c = \frac{[D]^d [E]^e [F]^f}{[A]^a [B]^b [C]^c} \dots\dots (4)$$

$$\text{or, } K_c = \frac{C_D^d \times C_E^e \times C_F^f}{C_A^a \times C_B^b \times C_C^c} \dots\dots (5)$$

$$\text{or, } K_p = \frac{P_D^d \times P_E^e \times P_F^f}{P_A^a \times P_B^b \times P_C^c} \dots\dots (6)$$

Now, we know from ideal gas's equation,

$$PV = nRT$$

$$\text{or, } P = \frac{n}{V} RT \quad [C = \frac{n}{V}]$$

$$\text{or, } P = CRT \dots\dots\dots (7)$$

Putting the value of 'P' in eqn (6), we get,

$$K_p = \frac{(CRT)_D^d \times (CRT)_E^e \times (CRT)_F^f}{(CRT)_A^a \times (CRT)_B^b \times (CRT)_C^c}$$

$$\text{or, } K_p = \frac{C_D^d \times (RT)^d \times C_E^e \times (RT)^e \times C_F^f \times (RT)^f}{C_A^a \times (RT)^a \times C_B^b \times (RT)^b \times C_C^c \times (RT)^c}$$

$$\text{or, } K_p = \frac{C_D^d \times C_E^e \times C_F^f \times (RT)^{d+e+f}}{C_A^a \times C_B^b \times C_C^c \times (RT)^{a+b+c}}$$

$$\text{or, } K_p = K_c \times (RT)^{(d+e+f) - (a+b+c)} \quad [\text{from eq}^n \text{ (5)}]$$

$$\text{or, } K_p = K_c \times RT^{(n_2 - n_1)}$$

$$\therefore K_p = K_c \times (RT)^{\Delta n}$$

Where, $\Delta n = \text{Total mole number of products}(n_2) - \text{total mole number of reactants}(n_1)$

✶ Le-Chatelier Principle:

"If a change occurs in one of the controlling factors, such as temperature, pressure, concentration etc. under which a system is in equilibrium the system will tend to adjust itself in such a way so as to reduce the effect of that change."

In other words,

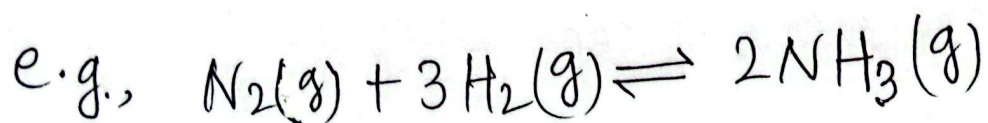
"When a chemical reaction at equilibrium is subjected to a stress, the system will shift the equilibrium point in the direction that counteracts the stress applied."

There are three ways in which the stress can be caused on a chemical equilibrium:

- ① Changing the concentration of a reactant or product.
 - ② Changing the pressure (or volume) of the system.
 - ③ Changing the temperature.
- ☞ If a change in concentration, pressure or temperature is caused to a chemical reaction in equilibrium, the equilibrium will shift to the right or the left so as to minimize the change.

✱ Effect of Change in Concentration:

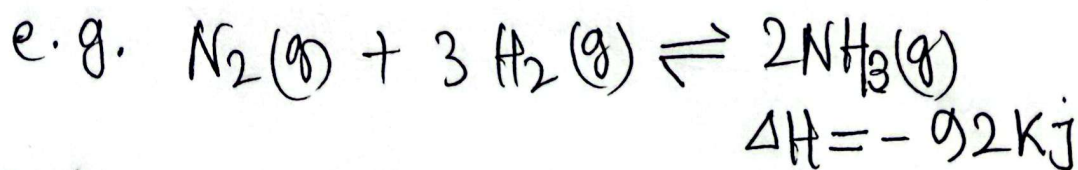
When concentration of any of the reactants or products is changed, the equilibrium shifts in a direction so as to reduce the change in concentration that was made.



When N_2 or H_2 is added to the eq^m stage, the equilibrium will shift to the right side so as to reduce the concentration of the N_2 or H_2 .

✱ Effect of Change of Temperature:

When temperature of a reaction is increased the eq^m shifts in a direction in which heat is absorbed.



It is an exothermic reaction and when

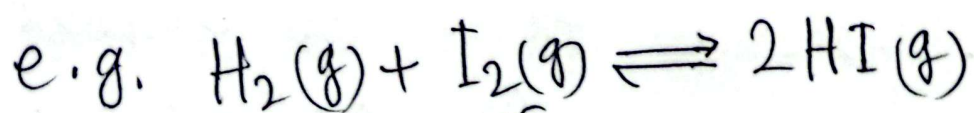
heat is increase in this reaction, the equilibrium will shift to the left.

Therefore, NH_3 Production will be decreased.

Effect of change of Pressure:

When pressure is increased on a gaseous equilibrium reaction, the equilibrium will shift in a direction which tends to decrease the pressure.

→ The chemical equilibrium in which no difference betⁿ total number of mole of reactants and total number of mole of products and the equilibrium isn't a gaseous equilibrium, there is no effect of pressure on equilibrium.



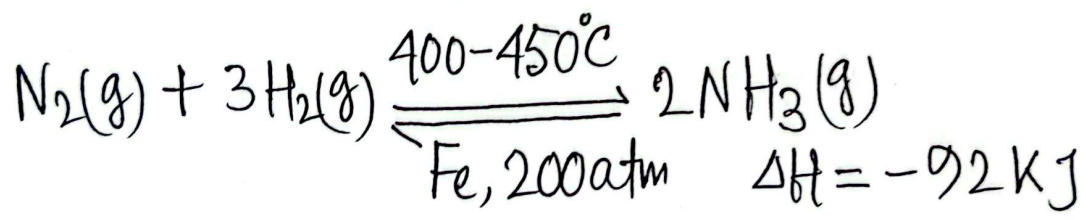
↳ No effect ^{of} pressure on equilibrium.

✳ Application of Le-Chatelier Principle:

1. Maximum Production Ammonia by Haber's Process:

A mixture of nitrogen and hydrogen gas in 1:3 ratio by volume is heated at a temperature of about $400-450^{\circ}\text{C}$ and at 200 atmospheric pressure in presence of 'Fe' (Iron) as catalyst to produce ammonia gas.

The reaction takes place as follows:



2. Maximum Production Sulphuric acid by Contact Process:

A mixture of oxygen and sulfur dioxide gas in 1:1 ratio by volume is heated

at a temperature of around 450°C and at 1-2 atm pressure in presence of V_2O_5 (Vanadium pentoxide) as catalyst to produce Sulfur trioxide.

The reaction takes as follows:

